CLASS 7 — SCIENCE · FORMULA SHEET

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | Quick revision, one page

Acids, bases and neutral substances

- Acids taste sour (lemon: citric acid; vinegar: acetic acid; curd: lactic acid; ant sting: formic acid).
- Bases taste bitter and feel soapy or slippery (soap solution, baking soda, lime water).
- Neutral substances (sugar solution, pure water) are neither acidic nor basic.
- Laboratory acids: hydrochloric acid HCl, sulphuric acid H_2SO_4, nitric acid HNO_3 (never taste them).

Indicators and their colour changes

- Litmus: acid turns blue litmus RED; base turns red litmus BLUE; neutral: no change.
- Turmeric: base turns it RED (reddish brown); acid and neutral: no change.
- China rose: acid gives dark pink (magenta); base gives green; neutral: no change.
- An indicator is a substance that shows by a colour change whether a substance is acidic or basic.

Neutralisation

- Acid + Base $\rightarrow$ Salt + Water (+ heat is released).
- Antacids (milk of magnesia) are bases that neutralise excess stomach acid.
- Baking soda or calamine soothes an ant sting (formic acid) or nettle sting.
- Acidic soil is treated with quicklime or slaked lime; acidic factory waste is treated with a base.

Handling safely

- Do not taste or touch unknown chemicals; use natural indicators for testing at home.
- Dilute the acid in water, never add water to a concentrated acid.

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